AWS

COMPLETE NOTES

(AWS)



BEGINNER TO ADVANCED

Compute	Storage	Database	Security	Monitoring	Networking
EC2, Lambda	S3, EBS	RDS, DynamoDB	IAM, VPC	CloudWatch	VPC, Route 53

SWIPE TO NEXT →

FOLLOW | SHARE | COMMENT | LIKE

AWS

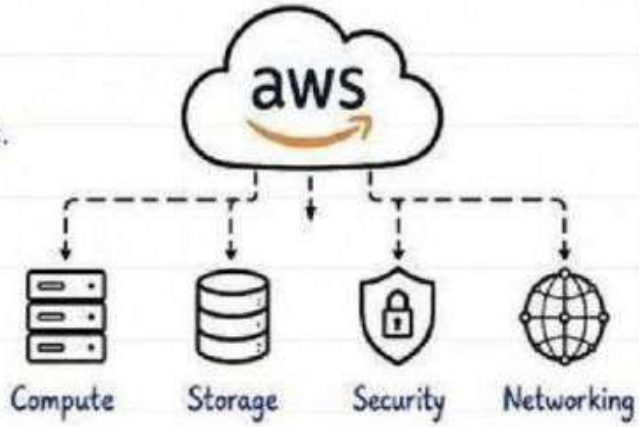
1

Amazon Web Services

World's most popular cloud platform by Amazon.
Build, deploy and scale applications on the cloud.

1 What is Cloud Computing?

- Delivery of IT services over the internet.
- On-demand resources, pay-as-you-go.
- No need to manage physical hardware.



2 What is AWS?

- AWS (Amazon Web Services) is a comprehensive cloud platform.
- Offers 200+ fully featured services.
- Global, reliable, secure and scalable.

3 Why Companies Use AWS?

- Reduce infrastructure cost
- Rapid innovation and agility
- Scalable and flexible resources
- High availability and reliability
- Global reach in minutes

4 Benefits of AWS

- Pay-as-you-go pricing
- Elasticity & Scalability
- High Availability
- Security & Compliance
- Wide range of services

AWS Global Infrastructure

- **Regions** – Separate geographic areas.
- **Availability Zones (AZ)** – Multiple data centers in each region.
- **Edge Locations** – CDN for fast content delivery.
- **Local Zones & Wavelength Zones** – Closer to users for low latency.
- **Multiple Data Centers** – Powering millions of applications worldwide.

Top AWS Services

EC2 S3 RDS Lambda VPC

REMEMBER

- You don't manage infrastructure.
- You focus on building the business.
- AWS handles the heavy lifting.

PRO TIP

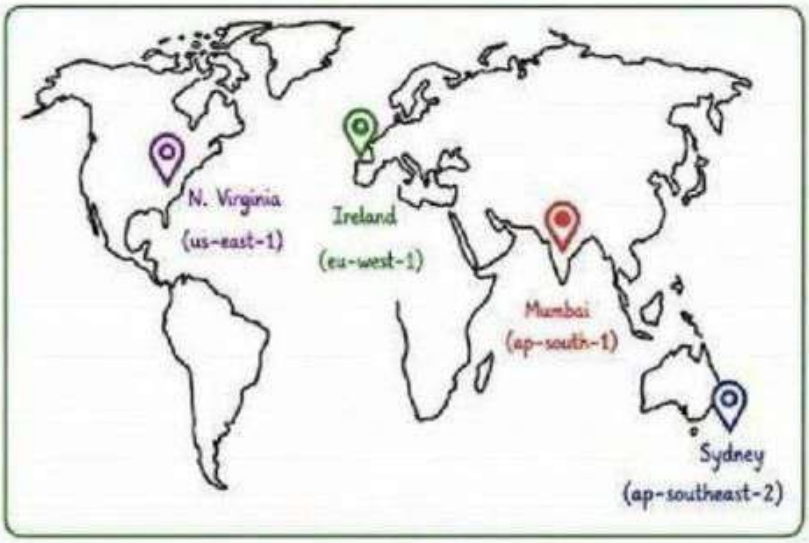
- Start Small, Scale as You Grow.
- Use Managed Services whenever possible.
- Monitor cost and optimize resources.

AWS Global Infrastructure

AWS offers a secure, reliable & global network of data centers to run your applications anywhere in the world.

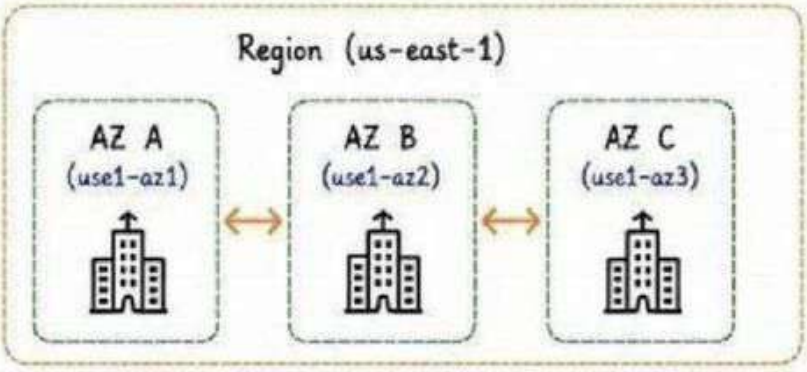
1 Regions

- A Region is a separate geographic area.
- Completely isolated from other Regions.
- Each Region has multiple AZs.
- Example: us-east-1 (N. Virginia)



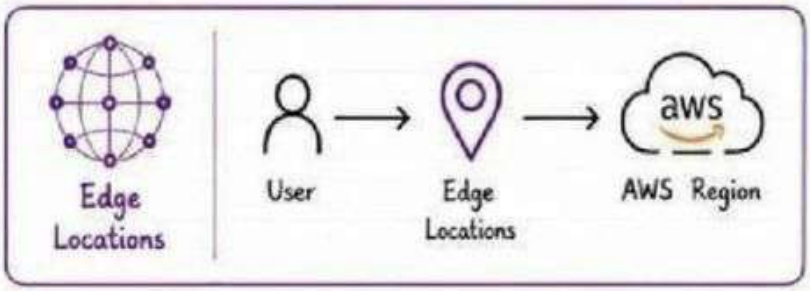
2 Availability Zones (AZ)

- AZs are one or more data centers with independent power, cooling and networking.
- Designed for high availability.
- AZs are connected with low latency.



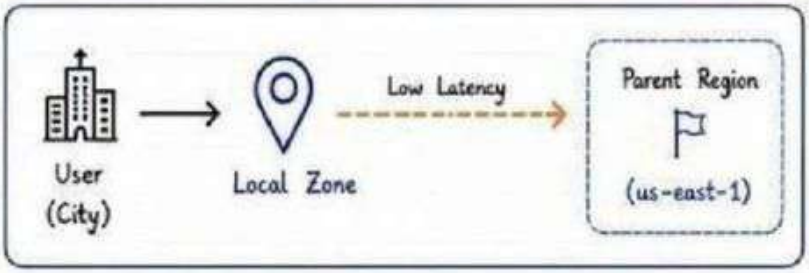
3 Edge Locations

- Used by CloudFront (CDN).
- Deliver content closer to users.
- Not used to run AWS resources.
- Hundreds of edge locations worldwide.



4 Local Zones

- Extend AWS infrastructure closer to large population centers.
- Lower latency for real-time apps.
- Dependent on a parent Region.



REMEMBER

- Region → Multiple AZs
- AZ → Multiple Data Centers
- Edge & Local Zones → Closer to Users

REAL-WORLD EXAMPLE

A user in India accesses your app.
Request goes to nearest Edge/Local Zone
Data is processed in the AWS Region (Mumbai).

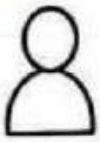
IAM (Identity & Access Management)

3

IAM helps you securely manage access to AWS services and resources for your users and applications.



1 Users



- An individual identity.
- Has unique credentials.
- Can belong to one or more groups.

2 Groups



- Collection of users.
- Permissions applied to the group.
- Easier management of access.

3 Roles



- Used by AWS services or external users.
- Temporary access.
- No long-term credentials.

4 Policies



- JSON documents.
- Define permissions (Allow / Deny).
- Attach to Users, Groups or Roles.

5 MFA (Multi-Factor Authentication)



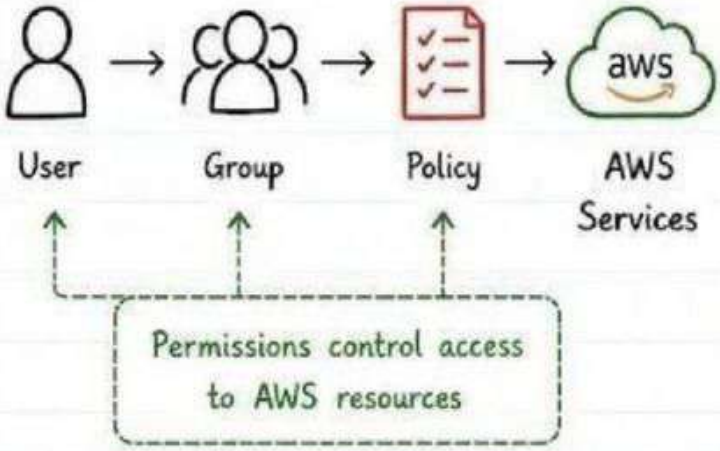
- Extra layer of security.
- Requires a second verification (OTP or Authenticator app).
- Protects against unauthorized access.

6 Least Privilege Principle



- Grant minimum permissions necessary.
- Reduces security risks.
- Best practice for all AWS accounts.

How IAM Works



Example Policy (Read Only S3 Access)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [ "s3:ListBucket", "s3:GetObject" ],
      "Resource": [
        "arn:aws:s3:::my-bucket",
        "arn:aws:s3:::my-bucket/*"
      ]
    }
  ]
}
```

REMEMBER



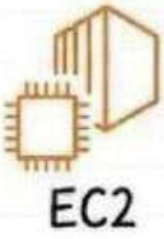
- Users are people.
- Roles are for applications or services.
- Policies define what is allowed.

BEST PRACTICE



- Use Groups to manage permissions.
- Enable MFA for all users.
- Follow Least Privilege always.

EC2 (Virtual Servers)



Amazon EC2 provides secure, resizable compute capacity in the cloud. You get virtual servers in minutes.

1 What is EC2?

- EC2 = Elastic Compute Cloud
- Virtual servers in AWS cloud.
- Full control of OS, storage, networking & security.

2 Instance Types

- Different types for different workloads.
- Example families: t, m, c, r, i, g
- Choose based on CPU, Memory, Storage, Network.

3 AMI (Amazon Machine Image)

- Pre-configured template.
- Includes OS, software & settings.
- Use AWS Marketplace, or custom AMI.

4 Key Pair

- Used to securely connect to EC2.
- Consists of Public key (.pem) & Private key.
- Keep private key safe!

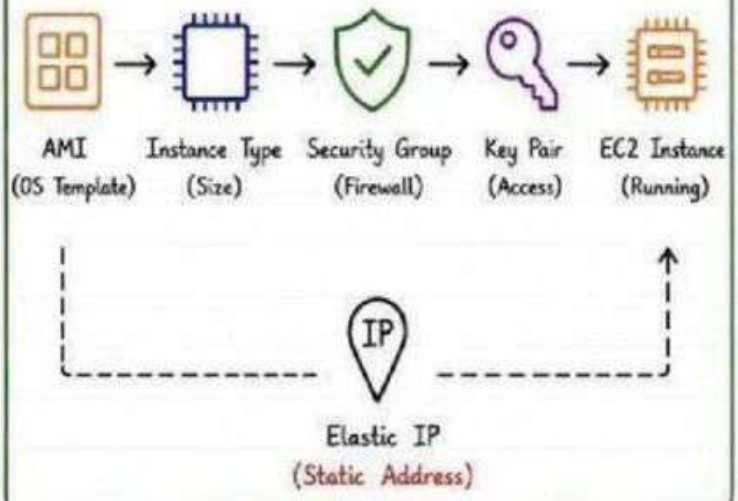
5 Security Groups

- Acts as a virtual firewall.
- Control inbound & outbound traffic.
- Rules based on IP, Port, Protocol.

6 Elastic IP

- Static IPv4 address.
- Not attached to instance by default.
- Re-attach to running instance.

EC2 Launch Flow



Example: Launch EC2 Instance

- [x] 1. Choose an AMI (e.g., Amazon Linux 2)
- [x] 2. Select Instance Type (e.g., t3.micro)
- [x] 3. Configure Security Group (Allow SSH 22)
- [x] 4. Create or Select Key Pair (.pem file)
- [x] 5. Launch Instance
- [x] 6. (Optional) Allocate & Associate Elastic IP

Connect to EC2 (Linux)

```
ssh -i your-key.pem ec2-user@<Public-IP>
```

REMEMBER

- You pay for running EC2 instances.
- Stop = Stop charging (EBS data safe).
- Terminate = Delete (EBS deleted if not attached).

BEST PRACTICE

- Use Least Privilege in Security Groups.
- Use IAM Roles for EC2 (not access keys).
- Enable monitoring with CloudWatch.
- Regularly stop unused instances.

Storage Services



AWS provides secure, durable and scalable storage solutions for every type of data.

① S3



- Object storage
- Highly scalable
- 99.999999999% durability
- For any type of data

② EBS



- Block storage
- For EC2 instances
- High performance
- Data tied to AZ

③ EFS



- File storage
- Shared across multiple instances
- Auto scaling
- Used with Linux

④ Glacier



- Archive storage
- Very low cost
- Data retrieval takes minutes-hours
- For backup & archive

Storage Comparison

Service	Type	Access	Durability	Performance	Use Case	Cost
S3	Object	Internet (HTTP/HTTPS)	11 9's (99.999999999%)	High	Static websites, backups, media, data lakes	$
EBS	Block	EC2 (Attached)	99.999%	High	OS, databases, applications requiring low latency	$$
EFS	File	NFS (Shared)	99.999%	Medium	Shared files, content management	$$
Glacier	Archive	Internet (Retrieval)	11 9's (99.999999999%)	Low	Long-term backup, compliance, archives	$

When to Use Which?

S3

Use for objects, backups, logs, static content, data lakes.

EBS

Use for EC2 boot volumes, databases, high performance apps.

EFS

Use for shared file systems across multiple EC2 instances.

Glacier

Use for long-term archives, compliance, cold data storage.

REMEMBER

- S3 = Object Storage (Internet scale)
- EBS = Block Storage (EC2)
- EFS = File Storage (Shared)
- Glacier = Archive Storage (Low cost)

BEST PRACTICE

- Choose the right storage for the right workload.
- Use Lifecycle policies in S3 to reduce costs.
- Encrypt data at rest and in transit.

Networking Basics

VPC

AWS Networking helps your resources communicate securely with the internet and each other.

1 VPC (Virtual Private Cloud)

- Your isolated network in AWS.
- Control IP ranges, subnets, route tables, gateways.
- Resources in a VPC can communicate privately.

2 Public vs Private Subnet

Public Subnet

- Has route to Internet Gateway
- Resources are publicly accessible

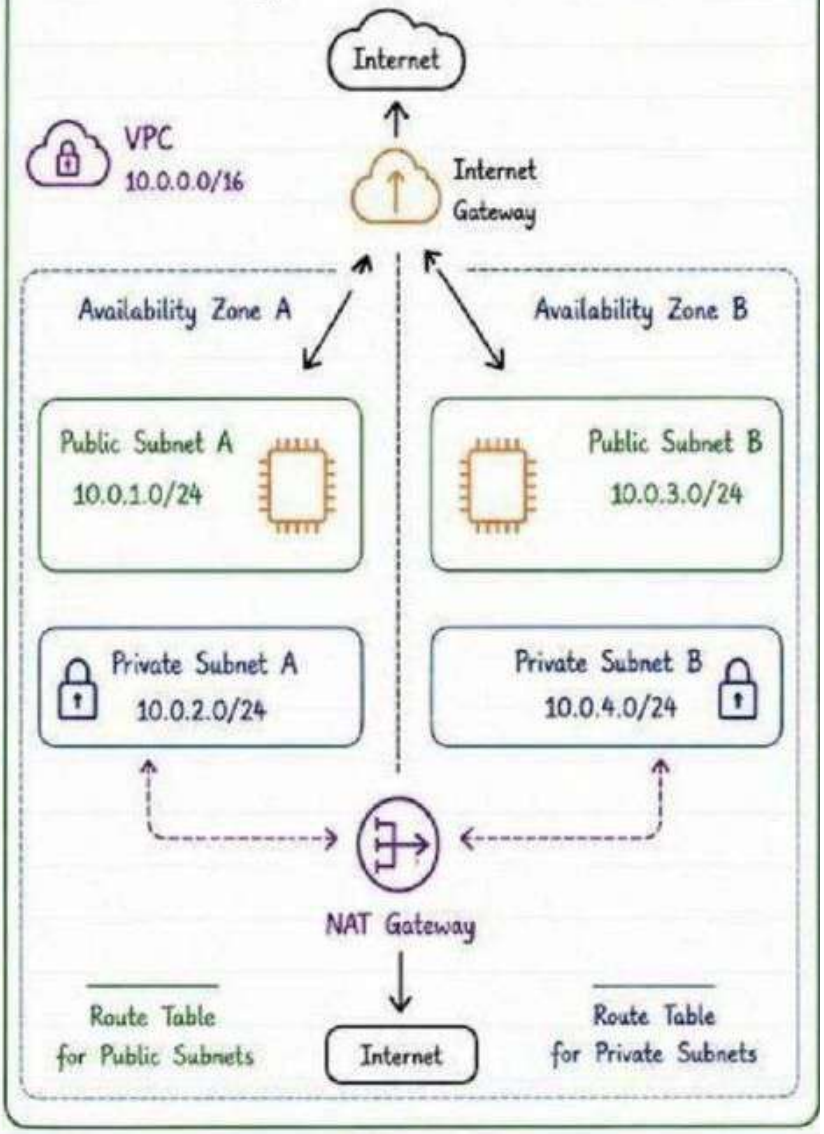
Private Subnet

- No direct route to Internet
- More secure and isolated

3 Internet Gateway (IGW)

- Connects your VPC to the internet.
- Enables internet access for resources in public subnets.

Typical VPC Architecture



4 Route Table

- Contains rules (routes) to direct traffic.
- Associated with a subnet.
- Decides where network traffic goes.

5 NAT Gateway

- Allows resources in private subnets to access the internet.
- Blocks inbound traffic from internet.
- One NAT Gateway per AZ for high availability.

6 CIDR Basics

192.168.0.0/16

- CIDR defines IP address range in a network.
- Format: A.B.C.D/n
- Example: 10.0.0.0/16 (65536 IPs)
- /24 = 256 IPs, /16 = 65536 IPs

REMEMBER

- Public Subnet → Internet access
- Private Subnet → No direct Internet
- NAT Gateway → Outbound Internet for Private Subnet

BEST PRACTICE

- Use Private Subnets for databases and backend services.
- Place NAT Gateway in each AZ.
- Use Route Tables wisely to control traffic flow.

7

Load Balancer & Auto Scaling

AWS

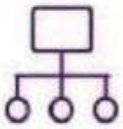
Distribute traffic, handle load & scale your applications automatically for high availability.

1 Why Load Balancer?



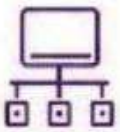
- Distribute incoming traffic across multiple instances.
- Prevent overload on a single server.
- Improve availability and fault tolerance.
- Perform health checks.

2 ALB (Application Load Balancer)



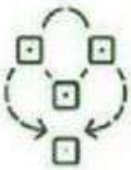
- Operates at Layer 7 (HTTP/HTTPS).
- Routes based on URL / Host / Path / Headers.
- Ideal for web applications and microservices.
- Supports SSL termination.

3 NLB (Network Load Load Balancer)



- Operates at Layer 4 (TCP/UDP).
- Handles millions of requests per second.
- Ultra low latency.
- Ideal for high performance applications.

4 Auto Scaling Group (ASG)



- Automatically adjusts the number of instances.
- Maintains application availability.
- Replace unhealthy instances automatically.
- Works with Load Balancers.

5 Scaling Policies

Target Tracking (Recommended)



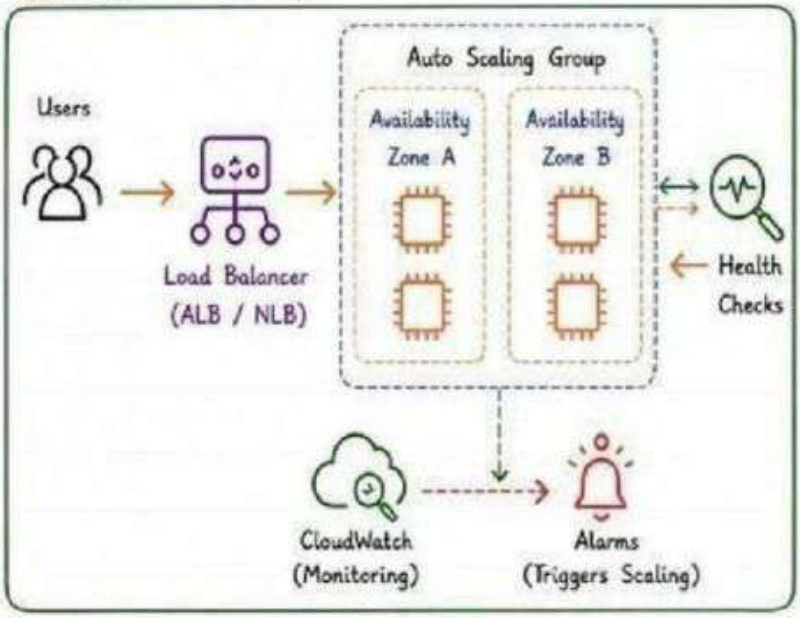
- Maintain a specific metric (e.g., CPU = 50%).
- Automatically adjusts capacity.

Step Scaling



- Scale based on CloudWatch alarms.
- Define steps for scale out or scale in.

6 High Availability



ALB vs NLB

Feature	ALB (Application)	NLB (Network)
Layer	Layer 7 (HTTP/HTTPS)	Layer 4 (TCP/UDP)
Use Case	Web apps, APIs, Microservices	High performance apps, Gaming, IoT
Routing	Path, Host, Query, Headers	IP, Port
Performance	Good	Ultra High
Static IP	No	Yes
SSL Termination	Yes	No (Pass-through)

REMEMBER

- ALB = Smart routing for applications.
- NLB = Speed & performance at network level.
- ASG = Scale out/in automatically.

BEST PRACTICE

- Use ALB for most web applications.
- Use NLB for high throughput & low latency workloads.
- Always enable Health Checks.
- Spread instances across multiple AZs.

Route 53 & DNS

Amazon Route 53 is a scalable and highly available DNS web service.

Route 53

1 DNS Basics

- DNS translates domain names to IP addresses.
- Works like the internet phonebook.
- Uses a distributed, hierarchical system.

2 Hosted Zone

- A container for DNS records.
- Public Hosted Zone (internet facing).
- Private Hosted Zone (inside VPC only).

3 Record Types

- A - Maps domain to IPv4.
- AAAA - Maps domain to IPv6.
- CNAME - Alias of another domain.
- MX - Mail exchange record.
- TXT - Text information.

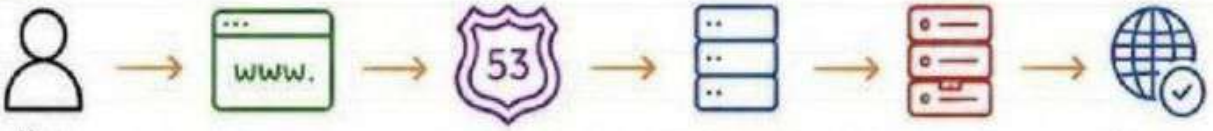
4 Health Checks

- Monitor the health of your resources.
- Checks via HTTP, HTTPS, TCP, or DNS.
- Used for failover routing.

5 Routing Policies

- Simple - Default, single resource.
- Weighted - Split traffic by weight.
- Latency - Route based on latency.
- Failover - Active-passive setup.
- Geolocation - Route by user location.
- Geoproximity - Route close to users.

6 Domain Flow



User Enters www.example.com → 1. Browser checks cache → 2. Query to Route 53 DNS → 3. Route 53 resolves IP → 4. Connection to the server → 5. Response to the user

Data is returned to the user

REMEMBER

- DNS = Domain Name System
- Route 53 = DNS service by AWS
- Converts domain names to IP addresses

BEST PRACTICE

- Use appropriate routing policy for your use case.
- Enable Health Checks for critical applications.
- Use Alias Records to route to AWS resources.